Mathematics: analysis and approaches
Standard level
Paper 1 — markscheme
2.1–2.5 Functions

1 November 2026

W4ss Academy practice set 1 | **Topic** 2.1–2.5 Functions | **Ref.** W4SS–MAA–T2–P1–01

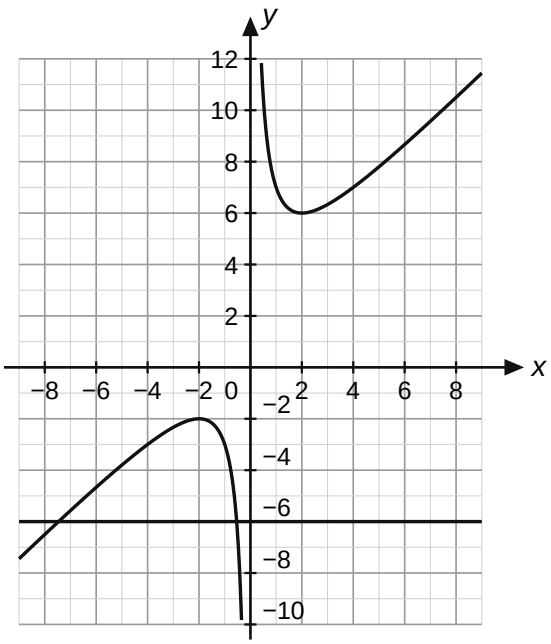
Student name

Paper 1

Instructions to students

- This markscheme is for teachers and for self-assessment after the test.
- **M**: method marks. **A**: answer or accuracy marks, usually after the M mark they depend on. **R**: reasoning marks. **N** (former course): marks for a correct answer with no working.
- Marks in brackets, e.g. **(M1)**, can be implied by correct working that follows. **AG**: answer given in the question, no mark for it. **FT**: follow through from an earlier error.
- The question export used for this paper has no official markscheme: these solutions were written by W4ss Academy in the IB style. Accept any correct equivalent method.
- Total: **[25 marks]**.

Section A

Question	Marks	Markscheme
1. (a)	**[1]**	$g(0) = -2$ ***A1***
1. (b)	**[2]**	attempt to find $f(g(0))$, e.g. $f(-2)$ ***(M1)*** $(f \circ g)(0) = 8$ ***A1***
1. (c)	**[2]**	recognizing from the table that $f(3) = 0$ ***(M1)*** $x = 3$ ***A1***
2. (a)	**[3]**	gradient of [BC] $= \frac{12-6}{-14-4} = -\frac{1}{3}$ ***(A1)*** use of $m_1 m_2 = -1$: gradient of $L = 3$ ***(M1)*** $y - 20 = 3(x + 2)$ OR $y = 3x + 26$ ***A1***
2. (b)	**[2]**	substituting $(k, 2)$ into their equation: $2 = 3k + 26$ ***(M1)*** $k = -8$ ***A1***
3. (a)(i)	**[1]**	horizontal straight line through $y = -6$, drawn across the whole grid ***A1*** 
3. (a)(ii)	**[1]**	2 (solutions) ***A1*** **Note:** the line meets the left branch twice and does not meet the right branch.

3. (b)	**[2]**	$-2 < k < 6$ ***A1A1*** **Note:** A1 for the end values −2 and 6, A1 for the strict inequalities. There is no solution for the values of k between the local maximum (−2) and the local minimum (6).
4. (a)	**[2]**	attempt to substitute $f(x)$ into g: $8 \times \frac{x+3}{4} + 5$ ***M1*** $= 2(x + 3) + 5$ ***A1*** $= 2x + 11$ ***AG***
4. (b)	**[3]**	**METHOD 1** $a = (g \circ f)(4)$ ***(M1)*** $= 2 \times 4 + 11$ ***(A1)*** $a = 19$ ***A1*** **METHOD 2** $(g \circ f)^{-1}(x) = \frac{x-11}{2}$ ***(M1)(A1)*** $\frac{a-11}{2} = 4 \Rightarrow a = 19$ ***A1***
5. (a)	**[2]**	$-11 \le f(x) \le 21$ ***A1A1*** **Note:** A1 for each correct end value with the correct inequality. Accept [−11, 21].
5. (b)	**[2]**	$x = 2y^3 + 5$ (interchanging x and y) ***M1*** $f^{-1}(x) = \sqrt[3]{((x-5)/2)}$ OR $((x-5)/2)^{1/3}$ ***A1***
5. (c)	**[2]**	domain: $-11 \le x \le 21$ ***A1*** range: $-2 \le f^{-1}(x) \le 2$ ***A1***

Please **do not** write on this page.

Answers written on this page
will not be marked.